The Reliability Tax: When Thinking Makes Language Models Less Trustworthy

Anonymous
Natural Language Processing, Course Project (Research Track)
anonymous@example.com

Abstract

Reasoning-augmented language models are deployed by default for high-stakes queries, and how long a model "thinks" is now a user-facing knob. The prevailing assumption is that more thinking yields more trustworthy answers. We test it. We formalize the capability–reliability trade-off as a measurable *reliability tax*, and ask whether it is a law—and whether it is *predictable*. A black-box budget sweep over four task types measures accuracy and three reliability axes jointly (calibration, selective prediction, abstention-versus-fabrication). A first real run already shows the tax is *task-dependent* and *switch-triggered*: on false-premise items, enabling thinking raises fabrication from 3.8% to 21.3%, while mathematical reasoning is unaffected.

1 Background & Motivation

Reasoning-augmented language models (LRMs) spend a variable amount of compute "thinking" before answering (DeepSeek-AI et al., 2025), and this budget is now a user-facing control (Muennighoff et al., 2025; Snell et al., 2024). The implicit contract is simple: *more thinking buys more trustworthy answers*. Practitioners act on it—raising the budget to "max" whenever an answer matters. Our survey (below) finds that contract contradicted across the field. If "think longer" is not a safe default, the budget must be chosen per query—which requires knowing whether the trade-off is *predictable*.

2 Research Questions & Hypotheses

RQ. Does increasing a model's thinking budget systematically trade reliability for accuracy, and can the degradation be foreseen? **H1 (task-dependence):** the ratio of reliability loss to accuracy gain differs across task types by at least $2\times$. **H2 (predictability, primary):** a predictor trained on a held-out half decides, at low budget and *without* seeing the high-budget answer, whether continuing will help, and the induced policy beats the best fixed budget in *usable accuracy* at matched cost. **H3 (negative control):** shuffling the trace features removes H2's advantage—if not, the predictor exploits an artefact. Each claim carries an explicit falsification condition, on a held-out prompt split.

3 Survey & Preliminary Literature

We surveyed the ACL 2026 landscape (1,409 papers) to locate the gap, and found the same warning recurring across otherwise unrelated subfields: capability gains are bought with reliability losses. Reinforcement learning that raises accuracy degrades calibration (Wang et al., 2026; Jin et al., 2025); reasoning enhancement amplifies tool hallucination, method-agnostically (Yin et al., 2025); "slower" models are less truthful under misleading input (Fang et al., 2025); reasoning models stay confidently wrong on false premises (Sathyanathan et al., 2026) and are *worse* at abstaining on unanswerable questions (Kirichenko et al., 2025); and they spend an order of magnitude more tokens while sometimes being *less* accurate (Srivastava et al., 2025; Shojaee et al., 2025).

This evidence is fragmented along three axes. It is *task-local* (one benchmark, often one model or recipe); *training-side* (the sharpest results concern what RLVR does to weights (Yue et al., 2025; Shao et al., 2025), which API users cannot reach); and *single-axis*—calibration *or* hallucination *or* abstention, never measured jointly, so we cannot tell whether the failures share a cause. The two closest studies vary the budget and measure one axis: Lacombe et al. (2025) show over-reasoning impairs calibration; Zhou et al. (2026) characterize its diminishing—sometimes negative—marginal utility. Both are *descriptive*: neither asks whether the losses across axes *co-move*, nor whether they

are *predictable* before a budget is committed. We therefore change the question from *whether* the trade-off exists to *whether it is a law*—and whether it can be turned into a deployment rule.

4 Method & Experimental Design

The object and the tax. Everything is read off the *budget–reliability profile*—accuracy and each reliability axis as a function of the thinking budget, reported against *realized* reasoning tokens since nominal knobs are not comparable across providers. With every score oriented so that higher is better, $\tau_r = (\partial \rho_r / \partial b)/(\partial \mathrm{acc}/\partial b)$ is the reliability given up per unit of accuracy gained; because τ_r diverges where accuracy saturates, our primary summaries are bounded instead—the fixed-coverage risk $R(c)$ and the area under the risk–coverage curve (Geifman and El-Yaniv, 2017).

Three axes. (i) *Calibration*: ECE (Guo et al., 2017) and verbalized overconfidence (Kadavath et al., 2022; Tian et al., 2023; Xiong et al., 2023). (ii) *Selective prediction*: $R(c)$ and AURC (Geifman and El-Yaniv, 2017; Wen et al., 2024). (iii) *Abstention vs. fabrication* on false-premise items, classified as refute / hedge / endorse, with fabrication rate = endorse share (Sathyanathan et al., 2026; Kirichenko et al., 2025).

Test sets. Four task types span where extra compute should help and where it can hurt, each with a verifiable gold standard:

- **Math reasoning** (GSM8K, MATH-500; numeric match) — the *positive control*, where thinking should help.
- **Factual QA** (Natural Questions; alias match) — closed-book recall, where thinking should mostly *waste* budget.
- **Multi-hop QA** (HotpotQA; alias match) — answers requiring composition of ≥ 2 facts; the reasoning test, where errors cascade.
- **False premise** — SciFact claims whose evidence *contradicts* them (Wadden et al., 2020), rewritten as questions that presuppose them. The only setting where the reliable behaviour is to *refuse*; our *negative control* and the sole source of the fabrication axis (classified refute / hedge / endorse).

200 items per task, with a prompt-level split held out for evaluating the stopping rule.

Budget and sampling. The budget axis uses the provider's official controls—thinking on/off and reasoning-effort levels—rather than `max_tokens`, which would confound budget with truncation; realized reasoning tokens are logged per response. Four budgets $\times$ two prompt templates $\times$ k=4 samples per item, with prompt-clustered bootstrap intervals (Zhang et al., 2025; Miller, 2024).

From profile to decision. The profile yields an oracle: the budget maximizing *usable accuracy*, i.e. accuracy at the coverage where ECE stays below a target ϵ. We then test whether that oracle is *findable* from low-budget features alone (realized length, verbalized confidence, self-consistency (Manakul et al., 2023), hedging markers), with the H3 permutation control.

5 Preliminary Result

A first real run on `deepseek-flash` (1,280 calls, 0 errors, ∼¥7–13) shows the tax is *task-dependent* and *switch-triggered*:

Task	Acc. off→max	ECE off→max	Fabrication off→max
Math	.988→.950	.01→.04	—
Factual QA	.475→.550	.45→.39	—
Multi-hop QA	.463→.562	.46→.41	—
False-premise	.962→.800	.03→.13	**.038→.213**

On false-premise items—where there is *no* accuracy to gain—enabling thinking multiplies fabrication by 5.6$\times$ and breaks calibration, while mathematical reasoning is untouched. Notably the effect is a *step* at the thinking on/off boundary rather than a smooth function of effort level, which sharpens H1 into a stronger, more falsifiable claim.

6 Expected Outcomes & Timeline

Outcomes. (1) A cross-task, cross-family test of whether the trade-off is a law, and where it is worst; (2) a held-out, cost-matched evaluation of an oracle-free *stopping rule*; (3) a released protocol, prompts and responses.

Timeline (single member; all roles by the author). **Wk 1–2** (done): survey, protocol and code, pipeline validation, first DeepSeek run. **Wk 3**: scale to 200 items $\times$ k=4; add Qwen3, GLM and Doubao. **Wk 4**: train and evaluate the stopping rule (H2/H3). **Wk 5**: final report, slides, artifact release.

References

DeepSeek-AI, Daya Guo, Dejian Yang, et al. 2025. Deepseek-r1: Incentivizing reasoning capability

in llms via reinforcement learning. *Preprint*, arXiv:2501.12948.

Sitong Fang, Wenjing Cao, Jiahao Li, et al. 2025. When slower isn't truer: Inverse scaling law of truthfulness in multimodal reasoning. *Preprint*, arXiv:2505.20214.

Yonatan Geifman and Ran El-Yaniv. 2017. Selective classification for deep neural networks. *Preprint*, arXiv:1705.08500.

Chuan Guo, Geoff Pleiss, Yu Sun, et al. 2017. On calibration of modern neural networks. *Preprint*, arXiv:1706.04599.

Renren Jin, Pengzhi Gao, Yuqi Ren, et al. 2025. Revisiting entropy in reinforcement learning for large reasoning models. *Preprint*, arXiv:2511.05993.

Saurav Kadavath, Tom Conerly, Amanda Askell, et al. 2022. Language models (mostly) know what they know. *Preprint*, arXiv:2207.05221.

Polina Kirichenko, Mark Ibrahim, Kamalika Chaudhuri, et al. 2025. Abstentionbench: Reasoning llms fail on unanswerable questions. *Preprint*, arXiv:2506.09038.

Romain Lacombe, Kerrie Wu, and Eddie Dilworth. 2025. Don't think twice! over-reasoning impairs confidence calibration. *Preprint*, arXiv:2508.15050.

Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. 2023. Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models. *Preprint*, arXiv:2303.08896.

Evan Miller. 2024. Adding error bars to evals: A statistical approach to language model evaluations. *Preprint*, arXiv:2411.00640.

Niklas Muennighoff, Zitong Yang, Weijia Shi, et al. 2025. s1: Simple test-time scaling. *Preprint*, arXiv:2501.19393.

Rose Sathyanathan, Kinshuk Vasisht, and Danish Pruthi. 2026. Evaluating reasoning models for queries with presuppositions. *Preprint*, arXiv:2605.03050.

Rulin Shao, Shuyue Stella Li, Rui Xin, et al. 2025. Spurious rewards: Rethinking training signals in rlvr. *Preprint*, arXiv:2506.10947.

Parshin Shojaee, Iman Mirzadeh, Keivan Alizadeh, et al. 2025. The illusion of thinking: Understanding the strengths and limitations of reasoning models via the lens of problem complexity. *Preprint*, arXiv:2506.06941.

Charlie Snell, Jaehoon Lee, Kelvin Xu, et al. 2024. Scaling llm test-time compute optimally can be more effective than scaling model parameters. *Preprint*, arXiv:2408.03314.

Gaurav Srivastava, Aafiya Hussain, Sriram Srinivasan, et al. 2025. Do llms overthink basic math reasoning? benchmarking the accuracy-efficiency tradeoff in language models. *Preprint*, arXiv:2507.04023.

Katherine Tian, Eric Mitchell, Allan Zhou, et al. 2023. Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. *Preprint*, arXiv:2305.14975.

David Wadden, Shanchuan Lin, Kyle Lo, et al. 2020. Fact or fiction: Verifying scientific claims. *Preprint*, arXiv:2004.14974.

Ziqi Wang, Xingzhou Lou, Meiqi Wu, et al. 2026. Calibration-aware policy optimization for reasoning llms. *Preprint*, arXiv:2604.12632.

Bingbing Wen, Jihan Yao, Shangbin Feng, et al. 2024. Know your limits: A survey of abstention in large language models. *Preprint*, arXiv:2407.18418.

Miao Xiong, Zhiyuan Hu, Xinyang Lu, et al. 2023. Can llms express their uncertainty? an empirical evaluation of confidence elicitation in llms. *Preprint*, arXiv:2306.13063.

Chenlong Yin, Zeyang Sha, Shiwen Cui, et al. 2025. The reasoning trap: How enhancing llm reasoning amplifies tool hallucination. *Preprint*, arXiv:2510.22977.

Yang Yue, Zhiqi Chen, Rui Lu, et al. 2025. Does reinforcement learning really incentivize reasoning capacity in llms beyond the base model? *Preprint*, arXiv:2504.13837.

Wenbo Zhang, Hengrui Cai, and Wenyu Chen. 2025. Beyond the singular: Revealing the value of multiple generations in benchmark evaluation. *Preprint*, arXiv:2502.08943.

Shu Zhou, Rui Ling, Junan Chen, et al. 2026. When more thinking hurts: Overthinking in llm test-time compute scaling. *Preprint*, arXiv:2604.10739.